

Project Management

THE ENVIRONMENTAL IMPACT OF GENERATIVE AI

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Context

Behind every AI request lies a resource-intensive process: servers compute, data centers consume electricity, cooling systems use water, and the resulting environmental footprint grows with AI usage.

**The question is :
How can public data help us estimate and communicate the environmental impact of generative AI?**

Four dimensions of our project

SCOPE — What will we do?

Investigate the environmental impact of generative AI, focusing the quantitative analysis on electricity consumption. Review existing research, analyse relevant historical data and explore future consumption through forecasts or scenarios. Produce visualisations and a final presentation. Clearly distinguish AI-specific consumption from broader data-centre consumption.

QUALITY — What makes the result good?

Use credible, documented sources. Check missing values, units and consistency. Clearly label measured and estimated values. Produce readable graphs with titles, units and sources. Test forecasts against historical observations and a simple baseline where feasible. Explain assumptions, uncertainty and limitations.

- A forecasting report (if the dataset is available).
- Charts illustrating trends or the current situation.

TIMING — When will we do it?

Follow the proposed six-week schedule: definition and literature review for session 2; dataset selection by the end of Week of session 2; cleaning in Week of session 3; historical analysis in Week of session 4; forecasts or scenarios in Week of session 5; final presentation in Week of session 6. Hold weekly progress meetings.

RESOURCES — What do we need?

Team members with assigned responsibilities; personal laptops; open datasets and research reports; Python or R for analysis; presentation software; a shared folder and task tracker; Notion to track and manage project progress; feedback from the teacher.

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We pitch our project



WHAT?

Assess the environmental impact of generative AI.

We focus on electricity consumption and its evolution over time, while considering broader impacts such as carbon emissions, water use and hardware production.

WHY?

Generative AI is growing rapidly, increasing pressure on energy and natural resources.

Understanding these impacts can support more informed and sustainable use of AI.

HOW?

Our further work. We will analyse relevant datasets, visualise historical trends and, where possible, develop forecasts or future consumption scenarios.

Stage	Tasks and activities	Expected output	Timing
1. Define the project	Agree on the research question, audience, scope and final deliverables. Assign team roles.	Short project description and responsibilities	Session 1
2. Review existing research	Read relevant studies. Identify the environmental indicators used and distinguish AI, data centres and broader digital activities.	Summary of findings and sources	Sessions 1–2
3. Find and assess datasets	Check variables, years, geographical coverage, units and whether values are measured or estimated. Assess suitability for forecasting.	Dataset inventory and choice of main dataset	Session 2
4. Prepare the data	Clean missing values and duplicates, standardise units and organise observations over time.	Clean dataset and record of changes	Session 3
5. Analyse historical trends	Plot consumption over time, calculate growth rates and compare relevant categories or locations. Investigate unusual changes.	Historical analysis with clear visualisations	Session 4
6. Explore future consumption	If the data are sufficient, compare forecasting methods and test them on recent observations. Otherwise, build scenarios with explicit assumptions.	Forecasts or scenarios, with uncertainty and limitations	Session 5
7. Interpret and present results	Explain what the findings tell us about the topic, identify limitations and prepare conclusions.	Final presentation, analysis notebook and supporting figures	Session 6
Ongoing: Manage the project	Hold a short weekly meeting, review progress, track problems and update responsibilities and deadlines.	Updated task board and meeting notes	Every week